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Sooner or Later? Promising Delivery Speed in Online Retail

Ruomeng Cui,^a Zhikun Lu,^{a,*} Tianshu Sun,^{b,c} Joseph M. Golden^d

^aGoizueta Business School, Emory University, Atlanta, Georgia 30322; ^bDepartment of Technology and Operations, Cheung Kong Graduate School of Business, Beijing 100738, China; ^cMarshall School of Business, University of Southern California, Los Angeles, California 90089;

^dPerfectRec.com, Seattle, Washington 98119

*Corresponding author

Contact: ruomeng.cui@emory.edu,  <https://orcid.org/0000-0003-0939-8759> (RC); zhikun.lu@emory.edu,  <https://orcid.org/0000-0002-4869-2099> (ZL); tianshusun@ckgsb.edu.cn,  <https://orcid.org/0000-0002-9786-044X> (TS); jgolden9@gmail.com,  <https://orcid.org/0000-0002-2068-3676> (JMG)

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Abstract. Problem definition: Online retailers have to provide customers with an estimate of how fast an order can be delivered before they decide to make the purchase. Retailers can strategically adjust this delivery speed promise online without changing offline infrastructure, and doing so may fundamentally impact business outcomes. It can influence consumers' purchasing decisions and postpurchase experiences, often in the opposite direction. On one hand, an aggressive (i.e., faster) delivery estimate could ensure that more customers meet their deadlines and thus, may increase their purchases ex ante. On the other hand, an aggressive estimate tends to over-promise, potentially leading to a longer than expected wait time, which can lower customer satisfaction and increase product returns ex post. In this research, we estimate the causal effect of retailers' delivery speed promise on customer behaviors and business performance. **Methodology/results:** Collaborating with [Collage.com](https://www.collage.com), an online retailer that sells customized photo products across the United States, we exogenously varied the disclosed delivery speed estimates online while keeping the physical delivery speed unchanged. Using the difference-in-differences identification strategy, we find that a faster promise increases sales and profits, but it also increases product returns and reduces customer retention. In addition, we propose a data-driven model that uses the estimated parameters to optimize delivery promises to maximize customer lifetime value. **Managerial implications:** Our findings provide managerial insights and a data-driven policy that retailers can leverage to optimize and customize their delivery promises.

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Keywords: delivery speed information • online retail • innovative business strategy

1. Introduction

Online retail is thriving, and it continues to expand at an impressive rate; e-commerce sales in the United States accounted for 10.7% of total retail sales in the first quarter of 2019 and are expected to grow three times faster than traditional retail (MarketingCharts 2019). Retailers show customers various types of information online about their products and services, such as prices, reviews, inventory availability, and delivery speed. How to effectively leverage such information in an innovative way has become an important decision for online retailers to manage the customer experience. For example, the delivery service is a critical decision factor in online shopping; shipping speed has been shown to affect the decisions of 87% of online shoppers, and speed is deemed more important than price when there is a deadline (Chao 2016, MH&L 2016). Because consumers increasingly value delivery, it is not only speed and reliability that are key issues in e-commerce; the way in which delivery information is disclosed to customers is also crucial for a winning strategy (Mottl 2018).

Retailers manage delivery from two perspectives: infrastructural and structural choices. After deciding on the level of delivery speed and reliability when serving customers, retailers can invest in infrastructure, such as facilities and transportation, to achieve the predetermined quality. In addition to the existing infrastructure, retailers may leverage structural choices, such as how to promise delivery speed, set shipping prices, and design shipping options for customers to further improve their experience. There is a vast amount of literature that has empirically evaluated infrastructural choices, including shortening the physical delivery time (Fisher et al. 2019), improving the fulfillment lead time (Lim et al. 2023), and having high-quality delivery options (Cui et al. 2020). However, the studies on structural choices are primarily theoretical, such as how to optimally design delivery time commitment and service capacity (So and Song 1998, Ho and Zheng 2004, Shang and Liu 2011, Wu et al. 2012) and how to present shipping prices (Gümüç et al. 2013). There is limited empirical evidence on structural choices, especially on how to manage delivery speed

information when keeping the offline infrastructure and physical delivery process unchanged. We contribute to the literature by empirically quantifying the information aspect of delivery speed and its impact.

In practice, the physical delivery speed is always subject to uncertainty because of production, inventory, scheduling, or transportation factors. For example, retailers may require a longer time to prepare packages because of seasonality, and carriers may wait for trucks to be fully loaded before shipping in large batches, which makes customers' wait time uncertain. In such circumstances, how to promise customers delivery speed becomes a strategic lever to manage their expectations, which in turn, determines their purchase decision and postpurchase responses. McKinsey has reported that many companies overlook consumer behavior when delivering services and recommends that companies pay attention to customer expectations (DeVine et al. 2012, Bhattacharjee et al. 2016). Therefore, it is important to understand consumers' reactions to delivery speed information, as they help retailers improve their information disclosure policies.

In this paper, we study how this innovative information strategy—leveraging and managing delivery speed information—creates value for online retailers. In particular, we study whether and to what extent consumers respond to online retailers' promised delivery speed. We examine how aggressive and conservative promises affect customers' purchasing, product return, and retention behaviors as well as companies' profitability.

An aggressive or conservative delivery promise has both benefits and costs. When a retailer chooses a more aggressive disclosure, customers expect a faster delivery. This faster promise can result in meeting more customers' deadlines and lead to more purchases. In contrast, a more conservative disclosure indicates a longer wait and stimulates fewer purchases. However, once customers have placed an order, they form expectations about how long they will have to wait based on the delivery information promised. When a retailer promises an aggressive delivery speed but keeps the actual speed unchanged, customers are likely to experience a longer than expected wait time. This might hurt their purchasing experience, cause more returns, and in the worst case, discourage future purchases. A conservative disclosure, however, might improve a customer's purchasing experience because their order might arrive sooner than expected. In short, retailers need to balance this trade-off between customers' expected and experienced delivery speed.

To test the effect of promised delivery speed, we partnered with [Collage.com](#), a leading online retailer that sells custom photo products in the United States. Collage enables customers to upload photos, design picture layouts, adjust backgrounds, and edit text to create custom products and gifts, such as mugs, blankets, pillows, and

canvases. The retailer contracts with dozens of vendors capable of producing customized items. It also partners with third-party carriers to deliver products from the vendors to customers. The company used to have a speed disclosure policy where customers across all cities received the same delivery estimates. In a 2016 internal email to the executives, the company's chief executive officer (CEO) expressed concern that their shipping rule was too inflexible and that he was thinking about making it more customized.

We were actually being inflexible in our initial shipping rules. We will make our estimates much more customized.—Joe Golden, Cofounder and CEO of [Collage.com](#)

We worked with Collage to design a new shipping disclosure policy that was implemented on December 9, 2016. Customers in different cities were given different delivery speed estimates, which would be faster in 13.5% of cities, slower in 58.5% of cities, and not change in the remaining 27.9% of cities. This created two groups of cities, one under the influence of the aggressive disclosure and one under the conservative disclosure, and a control group without changes in speed estimates. Because all shipping was outsourced to third-party carriers and because the retailer did not alter its physical delivery setup, this new policy only changed the delivery speed information online and consumers' expectations with regard to delivery speed. In this way, the policy change worked as a quasiexperiment and created exogenous variations to identify the causal effect of delivery speed promises.

We collected Collage's transaction-level data set of 175,647 transactions from 462 stock-keeping units (SKUs), which spanned from September 2016 to March 2017 (i.e., 12 weeks before and after the policy change). We also aggregated the data to create a panel structure across 7,090 cities over 24 weeks. The data include the standard order information (i.e., the purchase value of an order, the price of each purchased item, how much customers paid for shipping, and the profit margin of each order), how satisfied customers felt with the purchasing experience and delivery of their orders (i.e., satisfaction rate and score), and product return information (i.e., whether the product was returned). In this study, we assess five outcome variables that measure sales and expenditure—the number of orders, profit, order value, item price, and shipping expenditure. Additionally, we study three outcome variables that measure customers' postpurchase decisions—product return, satisfied or not, and satisfaction score. Further, we track customers' future transactions over a three-year period after our studied period to measure their retention behavior. We also collect each order's promised delivery speed (i.e., the promised delivery date minus the order date) and actual delivery speed (i.e., the actual delivery date minus the order date). We define the treatment variable as the number of days the

retailer promises a faster or slower speed under the new disclosure policy relative to the promise under the old disclosure policy, a similar measure to that employed by Yu et al. (2022).

We use the difference-in-differences (DID) identification strategy to study customers' purchasing decisions. We exploit the temporal and geographical variations in the market outcomes of the cities treated with the aggressive disclosure (i.e., cities that received a faster delivery estimate), the cities treated with the conservative disclosure (i.e., cities that received a slower delivery estimate), and untreated cities (i.e., cities that received an unchanged delivery estimate). The treatment reflects a continuous change in the delivery promise ranging from 5.5 days faster to 6 days slower. We verify the parallel trend assumption that the treatment and control groups followed the same trend before the policy change. In addition, we use transaction-level and customer-level regressions to estimate the effect on product returns, customer satisfaction, and customer retention.

We find that the change in the disclosure strategy significantly affects consumers' purchasing behavior. In particular, when the retailer promises customers one-day faster shipping, sales increase by 0.73%, profits increase by 2.0%, and customers spend 3.5% more on each order. This could indicate a significant business impact for the studied company; by varying the promised speed by only one day, Collage would receive 30,000 more orders and \$9 million more in revenue. However, when the retailer promises customers one-day slower shipping, sales decrease by 0.51%, profits decrease by 2.7%, and customers spend 3.1% less on each order. We also show that the aggressive disclosure increases product returns, whereas the conservative disclosure does not significantly affect returns. Moreover, the aggressive disclosure (because it causes more delays and product returns) lowers customer retention, whereas the conservative disclosure (because it leads to fewer delays) increases customer retention. These results indicate that, in our research context, promising customers a faster delivery speed can boost sales and profitability but at the cost of a higher product return rate and a lower retention rate. In addition, retailers should exercise caution when trying slower delivery promises because customers may make fewer purchases and buy cheaper products despite their higher retention rate.

Our findings indicate that customers are in fact sensitive to delivery speed promises when making purchases online. It is important to note that customers can decide on whether to purchase, to return products, and to revisit based on how retailers promise the delivery speed. If online retailers manage customers' expectations properly, they will be able to improve their profitability and customer retention without investing heavily in logistics infrastructure.

2. Literature Review

Our paper is related to the stream of literature on the value of service time, especially in terms of delivery speed and reliability. The literature shows that this topic has been studied from the physical perspective and the information perspective. Previous empirical studies have primarily examined the effects of physical delivery speed or quality: for example, speedy service achieved by increased bike accessibility in a bike-sharing system (Kabra et al. 2020), a faster drive-through at a fast-food restaurant (Allon et al. 2011), a shorter queue at a store (Lu et al. 2013), compensation promotions that reduce wait time (Cohen et al. 2022), and a higher service level (Fisher et al. 2006). Consumers' sensitivity to delivery time is shown to vary across online and offline channels (Lim et al. 2023). Recent works provide empirical evidence that a longer observable queue length (Batt and Terwiesch 2015) or a delay announcement (Akşin et al. 2016) can change how customers perceive service speed, thus leading to more customers abandoning the queue. The literature has also studied the impact of delivery quality on consumers' postpurchase decisions. For example, Rao et al. (2014) show that products are less likely to be returned when the delivery is reliable because retailers' promises of timeliness are in line with their actual performance. Musalem et al. (2017) provide evidence that customers' experience of shipping via a particular route affects their future shipping decisions not only about that route but also about other routes serviced by the same logistics company.

Service reliability or more specifically, on-time performance (OTP) also play an important role in the airline industry. For example, airlines with poor OTP are penalized in the financial market and receive lower equity valuations (Ramdas et al. 2013). To improve OTP, Arıkan et al. (2013) propose a data-driven model that optimizes airline scheduling. Robust scheduling has been shown empirically to reduce delays and increase revenue and profits (Deshpande and Arıkan 2012, Atkinson et al. 2016). Moreover, Manchiraju et al. (2021) demonstrate that besides schedule buffering, operational improvement and network changes are important drivers of OTP.

Our work is closely related to Fisher et al. (2019) and Cui et al. (2020), who showed the benefits of fast and high-quality delivery to retail platforms. For example, Fisher et al. (2019) demonstrated that when physical delivery speed becomes faster, total revenue increases by 4% and net profits increase by 2.2% for an online retailer. Cui et al. (2020) found that when a high-quality delivery option was removed from Alibaba's retail platform, sales dropped by 14%; when it was resumed, they went back to normal.

In the mentioned research, the studied companies changed their physical facility networks and thus, shortened or lengthened their actual delivery time. However,

our paper focuses on the information side of delivery decisions. The information strategy can be executed at almost zero cost, and at the same time, it can profoundly impact consumer choices. Delivery information strategies have been well studied in the theoretical literature. Theoretical frameworks have been developed to understand the interrelations among firms' strategic pricing, delivery time promises, and capacity allocation choices (So and Song 1998, Ho and Zheng 2004, Shang and Liu 2011, Wu et al. 2012). Their models capture the trade-off between the benefits of a tight delivery-time promise to attract customers and the resulting costs of a congested system. However, the literature lacks empirical evidence on the effects of retailers' delivery information strategies. Yu et al. (2022) study the role of dynamic wait time information in virtual queues on a ride-sharing platform. They focus on the connection between the wait time disclosure and the immediate customer demand; it is less clear if there are any long-term consequences of alternative disclosure policies in their context. In our context, however, the delivery speed disclosure policy affects not only the immediate customer demand but also the postpurchase decisions that are linked to future customer demand. Our research complements the literature by empirically examining an important but understudied dimension of delivery services, namely how to strategically promise delivery speed in online retail environments.

This paper sheds light on a key aspect of market control—information sharing—for retail platforms. Various kinds of information have been shown to be effective for boosting sales and profits in online retail, such as sharing timely price information with sellers (Parker et al. 2016), revealing inventory availability information (Gallino and Moreno 2014, Cui et al. 2019, Calvo et al. 2023), and showing transparent delivery stages for online purchases (Bray 2023). We complement the literature by showing that delivery speed information, when its trade-offs are carefully leveraged, can attract demand, encourage expenditure, and improve customer retention.

3. Hypothesis Development

Now, we theorize how consumers respond to delivery speed information when shopping online. When deciding whether to buy a product, customers often assess the delivery service: that is, how fast the retailer can deliver the product. Shipping speed is considered a key factor that affects consumer shopping decisions; sometimes, it is more important than price when there is a deadline, such as when buying a holiday gift.

The promised delivery speed and the actual delivery speed jointly determine the customer experience. The shipping speed promised to customers reflects how fast customers expect to receive the product in the purchasing phase based on which customers determine whether

to place an order. As a result, the delivery promise can affect market outcomes, such as sales, profit, expenditure, and the types of products purchased. After an order is placed, it will take some lead time to be delivered, which includes manufacturing, customizing, packaging, and transportation time. In practice, the lead time is always uncertain. Therefore, the actual arrival time is often not exactly what the seller promised to its customers, as it is sometimes faster and sometimes slower. The promised delivery speed, in turn, affects how customers perceive delivery quality in the delivery phase. Consequently, the delivery promise can affect market outcomes, such as customer satisfaction, product returns, and customer retention.

Online retailers can strategically choose to promise the delivery speed in a more aggressive or conservative way. An aggressive disclosure means promising customers a relatively faster shipping speed, and a conservative disclosure means promising customers a relatively slower shipping speed. Inspired by the frameworks of So and Song (1998) and Ho and Zheng (2004), we develop a theoretical model in Online Appendix A to help illustrate how customers react to the delivery speed information in their purchasing and returning behaviors. The model captures the key factors that determine consumer choices. Next, we discuss the idea behind the model.

Retailers' delivery speed information affects consumers' decisions in two stages: purchasing and returning. We first discuss customers' ex ante purchasing decisions. Before an order is placed, the retailer promises a delivery speed, and the customer will rely on that information to make their purchasing decision. The promised speed helps customers form expectations about how fast they will receive their orders. The more aggressively (conservatively) the retailer promises its delivery speed, the faster (slower) customers expect the delivery. Customers' key decision factor is whether the expected delivery speed can meet their required deadline. Customers gain a positive utility from receiving the purchased products before their deadline. If the retailer fails to meet the deadline, customers' utility becomes much lower. For customers with a relatively urgent deadline, a conservative shipping speed promise will discourage them from buying because they would expect a higher probability of not receiving their order by their deadline, which makes the purchase less attractive. If the expected delivery speed becomes faster, customers would believe that the order will arrive before their deadline, and the risk of late arrival will be reduced. This causes the expected gain to exceed the purchasing cost for marginal customers who are hesitating to place an order, which encourages more sales. To conclude, a fast delivery promise is likely to result in meeting more customers' deadlines, which will lead to more purchases.

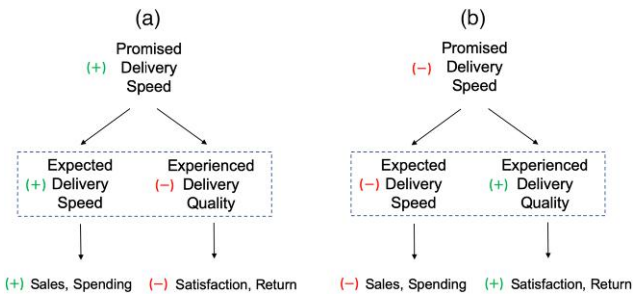
A retailer's speed promise strategy affects not only consumers' decision to buy or not but also their decision

about how much to spend on an order. Because a faster delivery promise effectively lowers consumers' utility loss associated with late arrivals, customers are willing to invest more in the purchase by upgrading to more expensive products; therefore, the aggressive disclosure can increase customers' spending on each order. Given that the shipping fee is proportional to the total spending, customers also spend more on shipping. In contrast, the conservative disclosure strategy informs customers of a longer delivery time, increasing the risk of late arrivals relative to customers' deadlines. Because late arrivals are more costly for customers when they buy expensive products, customers become more reluctant to upgrade their purchases and will spend less.

Hypothesis 1. *A faster delivery promise will increase the number of orders and customer expenditure per order. A slower delivery promise will reduce the number of orders and customer expenditure per order.*

In the purchasing stage, customers care about whether the promised delivery speed can meet their deadlines. In the returning stage, customers use the promised speed as the new benchmark in their timeline and care about retailers' ability to deliver an order by the promised date. Customers consider a delayed arrival beyond the promised wait time as a loss and an early arrival as a positive utility. When a retailer promises a more aggressive delivery speed while not changing the actual delivery speed, the seller may miss the promised delivery date. As a result, customers might not receive their orders on time, exert extra effort to track their packages, or even worse, miss their required deadline, making them unsatisfied and likely to return the product. Customers would blame the retailer for not fulfilling their promises, and in turn, they might even stop considering the seller a reliable retailer and in the worst case, no longer patronize it. A conservative disclosure, however, can improve a customer's experience because the order is more likely to arrive ahead of schedule. As a result, a company's ability to deliver faster than promised adds value to the purchasing experience, making customers more satisfied and less likely to return the product.

Figure 1. (Color online) The Research Framework



Notes. (a) Aggressive delivery speed promise. (b) Conservative delivery speed promise.

Hypothesis 2. *A faster delivery promise will increase product returns. A slower delivery promise will reduce product returns.*

We summarize our theoretical framework in Figure 1, which shows how delivery information strategies shape consumers' expected and experienced delivery speeds and thus, affect market outcomes. Figure 1(a) shows the impact of the aggressive disclosure, and Figure 1(b) shows the opposite impact for the conservative disclosure.

4. Research Background and Data

4.1. Research Background

We collaborated with a large national online retailer in the United States, Collage.com, that specializes in customized photo products. Customers can design a collage by uploading photos and customizing the layout with the proprietary software tools on the website. Once users create the layout, they can purchase various types of customized printed products, such as blankets, photo books, and canvases. The layout is then printed on physical materials, and the physical products are shipped to the customers. Every year, tens of millions of customers visit the retailer's website. Its annual revenue is more than \$100 million, with a transaction volume that exceeds millions of transactions per year. Given the volume of online traffic and sales, even a small improvement in the display of delivery speed information could lead to a large increase in the company's current and future sales.

The online retailer employs a business-to-consumer model and sells various types of customized prints directly to customers. The retailer contracts with dozens of vendors to manufacture the products. Once an order is placed on the website, the retailer assigns the order to a contracted vendor to manufacture it based on the customer's specifications. The retailer also partners with third-party carriers, including the U.S. Postal Service, UPS, DHL, and FedEx, to deliver products directly to customers after they are produced by the vendor. The retailer discloses the shipping speed estimates online for any products of interest and any cities via the interface shown in Figure 2. Because the production and shipping processes are managed in a centralized fashion, the retailer has full visibility of the actual production and shipping time for each order. Because of uncertainties in production and delivery, there are large variations in

Figure 2. (Color online) Disclosure of Delivery Speed Information

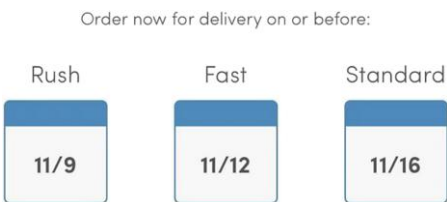
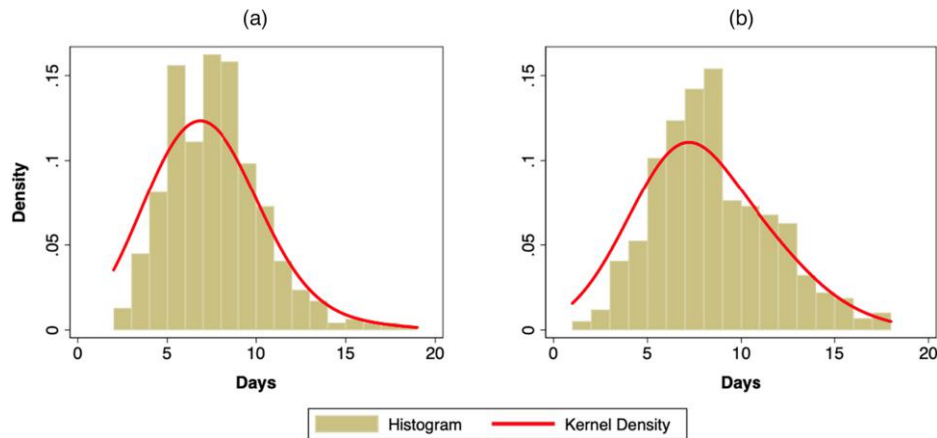


Figure 3. (Color online) Examples of Actual Delivery Speed

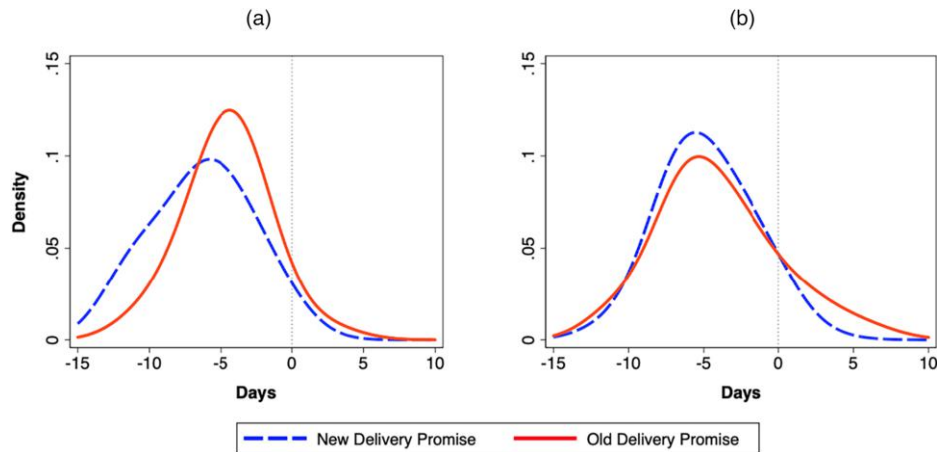
Notes. (a) Atlanta, Georgia. (b) Portland, Oregon. These two graphs show the distributions of the actual delivery speed in days at the city level. We present Atlanta and Portland as examples. The actual delivery speed is calculated by the actual delivery date minus the order date. As shown, even within the same city, the actual delivery speed varies.

the actual delivery speed; see Figure 3 for an example of the heterogeneity in the actual delivery time even within the same city. Therefore, customers will experience delays or early arrivals with any given delivery promise; see Figure 4 for an example. Given the range of actual delivery speed, retailers have a wide spectrum of shipping days from which to choose an estimate, and thus, there is room for them to improve the delivery speed promise.

The retailer used to have heuristics that disclosed the same delivery estimates for orders from the same city in the United States. For each city, the old estimate d^{old} equals the average delivery time in that city plus an adjustment term serving as a buffer to reduce delays. In the old policy, the average delivery time was computed

using limited historical data, and as a result, it was not accurate. The adjustment term was a rule-based constant dependent on the base delivery time. In 2016, Collage adopted a new delivery time promise d^{new} . They updated the average delivery time estimates with new data to obtain a more accurate estimate. In addition, they updated the adjustment term to be $\lambda\sigma_d$, where σ_d is the standard deviation of delivery time in a city and λ is a scalar used to control the delay rate.

At the same time, we aimed to study the causal impact of shipping speed promise. To do so, we followed Collage's new policy structure and properly introduced randomness into the speed promise change. Note that the treatment is the delivery promise change at the city level,

Figure 4. (Color online) Examples of Delivery Delay

Notes. (a) Atlanta, Georgia. (b) Portland, Oregon. These two graphs show the distributions of delivery delays based on the new and old delivery promises at the city level. We present Atlanta and Portland as examples. The delivery delay is calculated by the delivery date minus the promised date; thus, a negative number indicates the order was delivered earlier than promised, and a positive number indicates a delay. The delayed orders fall on the right side of the dotted line. As shown in the graphs, neither the old estimate nor the new estimate can accurately predict the actual delivery time, implying that considerable uncertainty exists in the delivery phase even within the same city. Moreover, it can be seen that the old estimate and the new estimate yield different delay distributions.

$\Delta d = d^{old} - d^{new}$. First, the promise change is primarily driven by newly added information, which has some randomness in nature. Thus, the promise can become either faster or slower. In other words, the construction of the new estimates helps the speed promise changes to be semirandom.¹ In addition, we adjusted the scalar λ to make sure the change in speed promise was uncorrelated with the previous actual shipping speed conditional on some city characteristics. We did this by adding some random noise to λ at the city level. In this way, the speed promise change would be exogenous to the pretreatment shipping speed. We empirically validate this condition in Section 8.3.

We implemented the new policy on December 9, 2016, and the policy change serves as a quasiexperiment and allows us to clearly identify the effect of delivery speed information.

4.2. Data Description

We collected a granular transaction-level data set for this study. First, for each order, we collected data on the purchase value, the price of each purchased item, the profit margin, how much customers paid for shipping, and the product return decision (i.e., whether the product was returned). Second, for each order, we collected the delivery estimate using the website’s event logs, and we also collected the actual delivery speed using shipping carriers’ tracking data. Last, we used postorder survey data to assess each customer’s satisfaction, including a binary “satisfied/not satisfied” response and a net promoter score (NPS) of 1–10. Table 1 presents the summary statistics.

Our data span from September 2016 to March 2017, including 175,647 transactions from 462 SKUs of photo products. We aggregated the data to create a panel structure over 7,090 cities across 24 weeks. We study three months before the policy change and three months after the policy change. Furthermore, we study five outcome variables that measure current sales and expenditure—total number of orders per city, total profit per city, purchase value per order, average item price per order, and shipping expenditure per order.² We study three outcome variables that measure customers’ postpurchase decisions that indicate their future retention—product return rate, satisfaction rate, and NPS satisfaction score. We also collect each order’s promised delivery speed (i.e., the promised delivery date minus the order date) and actual delivery speed (i.e., the actual delivery date minus the order date). We define the treatment variable as the number of days the retailer promised a faster or slower shipping speed under the new disclosure policy relative to the delivery speed promise under the old disclosure policy, $\Delta d = d^{old} - d^{new}$. For cities with $\Delta d > 0.5$, we define them as the fast group. For cities with $\Delta d < -0.5$, we define them as the slow group. For cities with $|\Delta d| \leq 0.5$, the actual impact is close to zero because of rounding, and hence, we define them as the control group.

We also collect customer-level data to study the impact of delivery delays. As suggested by the theory, aggressive (conservative) promises may bring negative (positive) shopping experiences because customers dislike delivery delays. Therefore, delivery delays would

Table 1. Summary Statistics

Variable	Cities with faster delivery promises		Control cities		Cities with slower delivery promises	
	960 cities		1,980 cities		4,150 cities	
	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
Before policy change						
Orders	0.60	1.93	1.21	4.28	1.21	4.05
Profit (\$)	49.41	160.59	99.02	363.37	95.13	316.60
Order value (\$)	106.29	56.15	107.03	122.61	103.08	55.48
Price (\$)	84.95	49.41	82.77	46.69	81.21	46.61
Shipping expenditure (\$)	13.16	5.69	13.13	7.37	12.96	5.98
Satisfied	0.93	0.25	0.91	0.26	0.89	0.30
NPS	9.12	1.96	9.05	1.94	8.83	2.20
Return rate (%)	0.35	5.19	0.52	5.91	0.52	5.74
After policy change						
Orders	0.55	1.71	1.11	4.00	1.05	3.60
Profit (\$)	52.77	168.07	99.98	372.78	90.71	433.42
Order value (\$)	125.87	97.66	116.65	80.17	108.68	63.45
Price (\$)	100.68	54.74	91.41	49.73	85.68	49.48
Shipping expenditure (\$)	15.56	8.88	14.60	7.44	13.93	7.38
Satisfied	0.92	0.26	0.93	0.25	0.92	0.26
NPS	9.16	1.87	9.13	1.84	9.06	1.92
Return rate (%)	0.60	6.95	0.54	5.89	0.60	6.32
Delivery promise change (days)	1.46	0.81	−0.07	0.32	−1.70	0.94

Table 2. Customer-Level Summary Statistics

Variable	Mean	Standard deviation	Observations
Orders	1.30	1.07	123,626
Order value (\$)	109.42	223.65	123,626
New customer (dummy)	0.79	0.40	123,626
Standard shipping	0.86	0.34	123,626
Delay rate (%)	2.58	15.25	123,626
Days of delay	0.09	0.75	123,626
Retention (dummy)	0.13	0.33	123,626
Return rate (%)	0.57	7.23	123,626
Satisfied	0.92	0.28	13,965
NPS	9.04	2.05	13,965

lower customer experience metrics, such as customer retention, product return, and customer satisfaction, all of which are statistically testable at the customer level. To test these predictions, we aggregate the transaction-level data to the customer level. Specifically, we first identify each customer who shopped at Collage during our sample period. For customers who placed multiple orders, we average the order features (e.g., order value and delay days) across their orders. Further, we track customers' future transactions over a one-year period. If the customer has made a transaction within one year after her last order in our sample period, the customer is marked as a retained customer, and the dummy indicator $Retention = 1$; otherwise, the customer is lost, and $Retention = 0$. The summary statistics are reported in Table 2.

5. Identification Strategy

Our primary goal is to identify the causal effect of faster or slower delivery speed information on consumer behavior, including both purchasing and postpurchasing decisions. Specifically, in Section 5.1, we leverage the exogenous policy change by using a DID regression to study purchasing decisions. In Section 5.2, we estimate the effect of delivery speed information on product returns and customer satisfaction using a transaction-level regression. Moreover, we conduct customer-level analysis in Section 5.3 to investigate how delivery delays may affect customer experience metrics, such as customer retention.

5.1. Difference-in-Differences Identification

To identify the effect of delivery on sales and expenditure, we estimate the following model specification with the city-level panel data,

$$Outcome_{it} = \alpha_i + \tau_t + \beta_1 Fast_i \times After_t + \beta_2 Slow_i \times After_t + e_{it}, \quad (1)$$

where i denotes the city and t denotes the week. To highlight the asymmetric impacts of faster and slower promises, we split our treatment Δd_i into two terms:

$$Fast_i = \begin{cases} \Delta d, & \text{if city } i \text{ in the fast group,} \\ 0, & \text{otherwise,} \end{cases}$$

$$Slow_i = \begin{cases} |\Delta d|, & \text{if city } i \text{ in the slow group,} \\ 0, & \text{otherwise.} \end{cases}$$

Here, $Fast_i$ is the continuous treatment of how many days faster the retailer promises delivery in city i , and $Slow_i$ is the continuous treatment of how many days slower the retailer promises. Variables α_i and τ_t are two fixed effects, where the former captures the time-invariant characteristics of city i and the latter captures the time-variant factors that are common to all cities. For the time fixed effect τ_t , we include a set of weekly time dummies that can flexibly control for nonlinear trends. Dummy variable $After_t$ indicates whether time t is after the policy change.³ Finally, $Outcome_{it}$ is the market outcome for city i in time t , which takes values in logged sales,⁴ profit,⁵ logged purchase value, logged price of purchased products, and logged expenditure in shipping. We validate the parallel trends assumption in Section 8.1.

5.2. Transaction-Level Analysis

To identify the effect of delivery on product returns and customer satisfaction, we estimate the following model specification at the individual transaction level:

$$Outcome_{ijt} = \alpha_i + \tau_t + \beta_1 Fast_j \times After_t + \beta_2 Slow_j \times After_t + \gamma X_j + e_{ijt}, \quad (2)$$

where j denotes the individual transaction, i denotes the city, and t represents the week since the policy change. We use $Outcome_{ijt}$ to represent individual-level outcomes, which takes values in the product return decision (dummy indicator), customer satisfaction (binary score), and NPS score (0–10). Again, $Fast_j$ is the continuous treatment of how many days faster the retailer promises delivery for transaction j , and $Slow_j$ is the continuous treatment of how many days slower the retailer promises delivery. If transaction j was made before the policy change, then $Fast_j = Slow_j = 0$. We also include two sets of dummy variables, α_i and τ_t , to incorporate the region fixed effect and the week fixed effect. Finally, we add control variables, X_j , to the regressions, which include the order value, whether a coupon is used, whether the customer is a new or existing customer, the shipping method, and the product type (i.e., SKU fixed effect).

In our data set, returns are rare events; only a small portion of customers disclosed their satisfaction. This low-frequency problem is further exacerbated when we aggregate the individual-level data into the city-level data, where variations at the individual level are smoothed out. Thus, the analysis at the aggregate level fails to capture the subtle and useful information hidden

Table 3. Impact of Delivery Speed Promise on Sales and Spending

	Log orders (1)	Profit (2)	Log order value (3)	Log item price (4)	Log shipping expense (5)
<i>Fast × After</i>	0.0073** (0.0030)	176.7* (94.7)	0.0347*** (0.0125)	0.0316** (0.0148)	0.0200* (0.0108)
<i>Slow × After</i>	−0.0051*** (0.0018)	−245.9*** (56.5)	−0.0310*** (0.0053)	−0.0436*** (0.0067)	−0.0313*** (0.0044)
City FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Observations	170,160	170,160	60,748	60,748	60,748
R ²	0.243	0.093	0.021	0.015	0.047

Notes. This table reports the estimated coefficients and cluster-robust standard errors (in parentheses) in Equation (1). The coefficients for the number of orders, profit, the purchase value per order, the average price per item, and consumers’ expenditure in shipping are presented in columns (1)–(5), respectively. The standard errors are clustered at the city level. FE stands for fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

in the data. To solve this issue, we employ Equation (2) to estimate the treatment effect on product returns and satisfaction at the individual transaction level.

5.3. Customer-Level Analysis

To study the impact of delays on customer retention, product returns, and customer satisfaction, we estimate the following model specification at the customer level:

$$Outcome_k = c + \beta Delay_k + \gamma X_k + e_k, \tag{3}$$

where k denotes the customer. For the independent variable $Delay_k$, we employ two different measures: the delay rate and the average days of delay across all customer k ’s orders during our sample period from September 2016 to March 2017. For the outcome variable, we use the retention dummy indicator (i.e., whether customer k revisits and makes any future purchases in the next year), the return rate, and the satisfaction rate.⁶ We also add control variables, X_k , to the regressions,

which capture a customer’s spending behaviors, including the number of orders, the average order value, whether the customer is a new or existing customer, the shipping method (i.e., the percentage of orders using the standard rather than premium shipping method), and city-level fixed effects.

6. Estimation Results
6.1. Impact of Delivery Speed Promise

We analyze customers’ responses when the promised delivery estimates changed. Table 3 reports the average treatment effect of delivery speed estimates on sales, profit, order value, the price of purchased items, and expenditure on shipping using Equation (1).⁷ Table 4 reports the average treatment effect of delivery speed estimates on product returns and customer satisfaction using Equation (2).

We find a positive treatment effect for the faster delivery promise and a negative treatment effect for the

Table 4. Impact of Delivery Speed Promise on Returns and Satisfaction

	Return		Satisfied		NPS
	(1) Linear	(2) Logit	(3) Linear	(4) Logit	(5) Linear
<i>Fast × After</i>	0.0006** (0.0003)	0.1013*** (0.0303)	−0.0025 (0.0026)	0.0140 (0.0346)	−0.0050 (0.0191)
<i>Slow × After</i>	−0.0004** (0.0002)	−0.0804** (0.0338)	0.0010 (0.0033)	−0.0461 (0.0305)	−0.0007 (0.0241)
Controls	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
Observations	175,647	170,554	13,719	15,635	13,719
R ²	0.012	0.055	0.059	0.076	0.077

Notes. This table reports the estimated coefficients and standard errors (in parentheses) in Equation (2). Regressions are run at the transaction level. The estimates for whether the order is returned using linear and logit models are presented in columns (1) and (2). The estimates for whether the customer is satisfied using linear and logit models are presented in columns (3) and (4). The estimates for the satisfaction score are presented in column (5). For linear models, city and week fixed effects are included, and standard errors are clustered at the city level. For nonlinear models, we use state-level fixed effects instead to avoid identification issues caused by the high dimensionality of city-level fixed effects. We report adjusted R^2 for linear models and pseudo- R^2 for nonlinear models. FE stands for fixed effect.

** $p < 0.05$; *** $p < 0.01$.

slower delivery promise on sales, which suggest that delivery information directly influences customers' purchase decisions. Column (1) of Table 3 shows that sales increase by 0.73%⁸ with a one-day faster promise but decrease by 0.51% with a one-day slower promise. This result confirms our hypothesis that a faster delivery speed promised to customers will lead to more purchases. Additionally, a one-day faster promise increases profits by \$1.77 or 2.0% per order, whereas a one-day slower promise decreases profits by \$2.46 or 2.7% per order.⁹ Further, an aggressive disclosure contributes to revenue not only through higher order volumes but also through greater order values. The order value increases at a rate of 3.5% per day with a faster delivery promise; however, it decreases at a rate of 3.1% per day with a slower delivery promise. Moreover, we identify that most changes in order values are because of changes in the price of the item purchased. A one-day faster delivery promise increases the average price of purchased items by 3.2% and increases consumers' shipping expenditure by 2.0%; in contrast, a one-day slower promise decreases these two categories by 4.3% and 3.1%, respectively. Therefore, Hypothesis 1 is supported.

However, there are consequences to faster delivery promises.¹⁰ We find a positive and significant treatment effect of a faster delivery promise on product return. As shown in column (1) of Table 4, a one-day-faster promise increases the return probability by 0.06 percentage points according to the linear regression model. This amount translates into a 9.7% increase in percentage given that the overall return rate in this subsample is 0.62%. In contrast, a one-day-slower promise decreases the return probability by 0.04 percentage points, which leads to a 6.4% reduction in product returns.¹¹ We also applied the probit and logit models, which yield very similar estimates. These results verify the concern that, without changing the physical delivery, an aggressive (conservative) promise causes more (less) delivery delays and

thus, leads to higher (lower) returns. Therefore, Hypothesis 2 is supported.

We also test the impact on customer satisfaction, which is shown in Table 4. Here, we have two different indicators that reflect customer satisfaction, *Satisfied* and *NPS*. We do not find statistically significant results once we control for order-level characteristics. These results should be interpreted with caution; only 8.5% of total customers disclosed their levels of satisfaction. This low response rate could have caused a sample selection problem, resulting in potentially biased estimates.

6.2. Impact of Delivery Delays

Next, we explore how delivery delays affect customer experience metrics by estimating Equation (3). We use two explanatory variables to capture different aspects of delays; *Delay rate* measures the percentage of orders that get delayed for a customer, and *Days of delay* measures the average days of delay experienced by a customer. Table 5 reports the estimation results.

We first study how delays impact customer retention. As shown in columns (1) and (2) of Table 5, both coefficients are negative and statistically significant, suggesting that delays hurt customer retention, and the longer the delays are that customers experience, the fewer customers are retained. During our sample period, the average number of orders is 1.3 per customer; most customers had only one order. For those customers, if their orders were delayed, then the revisit probability in the next year would drop by 1.29 percentage points according to our estimates, and for each additional delay day, the probability would further drop by 0.25 percentage points.¹² Therefore, our results indicate that an aggressive delivery speed promise lowers customer retention, and a conservative delivery speed promise improves customer retention via the impact on the level of delays experienced by customers.

Table 5. Impact of Delays on Customer Postpurchase Decisions

	<i>Retention</i> (1)	<i>Retention</i> (2)	<i>Return</i> (3)	<i>Return</i> (4)	<i>Satisfaction</i> (5)	<i>Satisfaction</i> (6)
<i>Delay rate</i>	-0.0129** (0.0058)		0.0288*** (0.0033)		-0.1894*** (0.0295)	
<i>Days of delay</i>		-0.0025** (0.0012)		0.0058*** (0.0008)		-0.0398*** (0.0071)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	123,626	123,626	123,626	123,626	11,861	11,861
R ²	0.0630	0.0630	0.0088	0.0087	0.0251	0.0267

Notes. This table reports the estimated coefficients and standard errors (in parentheses) in Equation (3) using linear regression. We estimate how returns and delays affect customer retention, return rate, and satisfaction rate. The outcome variable *Retention* is a dummy indicator (whether a customer revisits and makes any future purchases in the next year). One may also use the logit or probit model for *Retention*, which yields similar results. *Return* and *Satisfaction* are customer-level return rate and satisfaction rate, respectively. FE stands for fixed effect. The standard errors are clustered at the city level.

** $p < 0.05$; *** $p < 0.01$.

Next, we investigate how delays impact product returns and customer satisfaction. We again estimate Equation (3) using return rate and satisfaction rate as the outcome variables, and we find that delays lead to more returns and less satisfaction. We also perform this part of the analysis at the transaction level, and the results are highly robust.

7. Optimal Delivery Speed Promise

Our empirical analysis shows that a faster promise increases sales and order value at the cost of a higher product return rate and a lower retention rate. In other words, a faster promise boosts the retailer's immediate financial gains but causes a future financial loss. However, a slower promise does the opposite. These results suggest a trade-off between the short-term and long-term gains. By balancing the short-term gain and the long-term cost, the retailer can find the optimal promise that maximizes the total profitability. We now design a data-driven model that leverages the estimated customer responses to determine the best delivery speed promise.

We develop a model-based optimization framework. The model captures the drivers of profitability and how they depend on the promised speed. Next, we outline the ideas of the framework. The objective of a company is to maximize the customer lifetime value, which is the sum of a customer's discounted profit over time:

$$\text{Lifetime Value} = \pi + \beta\gamma\pi + (\beta\gamma)^2\pi + \dots = \frac{\pi}{1 - \beta\gamma},$$

where π is the average profit contributed by a customer in each period, β is the time discount factor that represents the cash flow discount and the natural growth of consumer spending, and γ is the retention rate that represents the probability that the customer will be retained in the next period. In our analysis, we use one year as our time period.

Further, we assume the annual profit as $\pi = \phi \times \text{Orders} \times \text{Value} \times (1 - \text{Return})$, where ϕ is the average profit margin, Orders is the average number of orders placed by a customer, Value is the average expenditure per order, and Return is the average return probability of each order. The latter three terms are functions of delivery speed promise t_p . Therefore, we can write the profit as

$$\pi(t_p) = \phi \times \text{Orders}(t_p) \times \text{Value}(t_p) \times (1 - \text{Return}(t_p)).$$

Moreover, because delivery speed promise changes delivery delays and delivery delays lower customer retention, we can also write the retention rate as a function of t_p : that is, $\gamma(t_p) = \tilde{\gamma}(\text{Delay}(t_p))$.

We have shown that the lifetime value of a customer can be represented as a function of delivery speed promise t_p , $\text{Lifetime Value}(t_p) = \pi(t_p)/(1 - \beta\gamma(t_p))$. To find the optimal t_p , we take the derivative of Lifetime Value with

respect to t_p , $\frac{\partial \text{Lifetime Value}/\text{Lifetime Value}}{\partial t_p} = \frac{\partial \pi/\pi}{\partial t_p} + \frac{\beta}{1 - \beta\gamma} \frac{\partial \gamma}{\partial t_p}$. We can see that when changing delivery speed promise, the change in customer lifetime value can be decomposed into two components. The first component $\frac{\partial \pi/\pi}{\partial t_p}$ captures the fact that a customer will spend more (less) when promised a faster (slower) delivery speed. The second component $\frac{\beta}{1 - \beta\gamma} \frac{\partial \gamma}{\partial t_p}$ captures the fact that customers are less (more) likely to be retained when promised a faster (slower) delivery speed. We further have $\frac{\partial \pi/\pi}{\partial t_p} = \frac{\partial \log(\text{Value})}{\partial t_p} + \frac{\partial \log(\text{Orders})}{\partial t_p} - \frac{\partial \text{Return}/\partial t_p}{1 - \text{Return}}$ and $\frac{\partial \gamma}{\partial t_p} = \frac{\partial \tilde{\gamma}}{\partial \text{Delay}} \frac{\partial \text{Delay}}{\partial t_p}$.

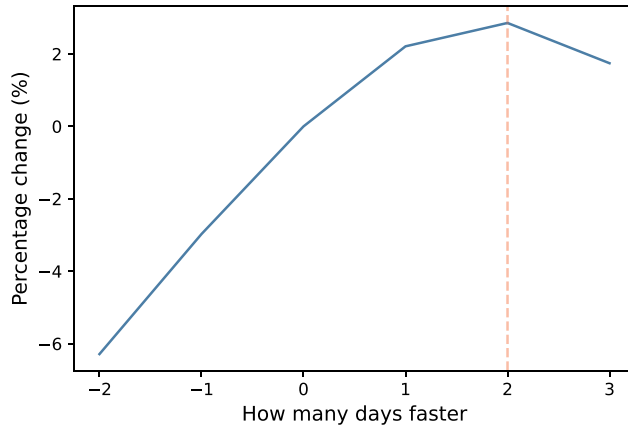
We next evaluate the impact of a change in delivery speed promise Δt_p on the percentage change of customer lifetime value using the following first-order approximation:

$$\begin{aligned} & \frac{\Delta \text{Lifetime Value}}{\text{Lifetime Value}} \\ &= \left(\frac{\partial \log(\text{Value})}{\partial t_p} + \frac{\partial \log(\text{Orders})}{\partial t_p} - \frac{\partial \text{Return}/\partial t_p}{1 - \text{Return}} \right) \Delta t_p \\ &+ \eta \frac{\partial \tilde{\gamma}}{\partial \text{Delay}} \Delta \text{Delay}, \end{aligned}$$

where $\eta = \beta/(1 - \beta\gamma) = \beta(1 + \beta\gamma + \beta^2\gamma^2 + \dots)$ captures a customer's future value conditional on being a retained customer in the next period relative to the customer's current value. According to the estimation from our data, the retention rate $\gamma = 89.25\%$. According to Collage's internal evaluation of β , η is set to be close to 10, which means that the total profit brought by an average customer in the future is 10 times as large as that of the current year conditional on being a retained customer in the next year. The parameters $\frac{\partial \log(\text{Value})}{\partial t_p}$, $\frac{\partial \log(\text{Orders})}{\partial t_p}$, $\frac{\partial \text{Return}}{\partial t_p}$, and $\frac{\partial \tilde{\gamma}}{\partial \text{Delay}}$ have been previously estimated. For a given Δt_p , we can directly observe the change in the delay rate ΔDelay using the order-level delivery records, which is plotted in Figure 10 in Online Appendix C.

The main results are displayed in Figure 5, which shows how customer lifetime value changes when the delivery promise becomes N day(s) faster than what the current policy assigns. When the retailer promises the delivery one day faster than what is currently promised ($n=1$), their customers' long-term profitability increases by 2.2%; when the promise becomes two days faster ($n=2$), the gain increases to 2.9%. Promising too aggressively beyond two days faster would start to decrease the gain. Thus, our analysis shows that the optimal promise for this retailer is on average two days faster than the status quo. By adopting a more aggressive disclosure policy, Collage can increase their customer lifetime value by close to 3%.

We perform a series of sensitivity analyses. First, we consider a situation where Collage's η equals 20, which means that the total profit brought by an average customer

Figure 5. (Color online) How Delivery Speed Promise Affects Customer Lifetime Value

Notes. This graph shows how customer lifetime value changes when the promise becomes N day(s) faster than the status quo. When $n = 1$, customer lifetime value increases by 2.2%; when increasing N to two, the gain in customer lifetime value increases to 2.9%, which is the maximum gain. Thus, the optimal delivery speed promise is two days faster than the status quo. By adopting a more aggressive disclosure policy, customer lifetime value can increase by close to 3%.

in the future is 20 times as large as that of the current year conditional on being a retained customer in the next year. If this is the case, the retailer faces a higher cost to delay, upset, and lose customers when promising customers a faster delivery speed. As a result, the retailer has less incentive to make an aggressive promise. Consistent with this idea, our optimization analysis presented in Figure 11 in Online Appendix C shows that the optimal promise is one day faster than the current disclosure policy. In addition, we report the optimal promise as a function of η and the associated maximum customer lifetime value in Figure 12 in Online Appendix C. As η becomes larger, customer future lifetime value weighs more, and hence, the optimal promise becomes less aggressive.

8. Robustness Check

In this section, we report several robustness tests to validate our results. Specifically, we test the parallel trends assumption, we show that the change in the delivery promise is independent of any potential changes in the physical delivery speed, and we validate the condition of our empirical design that the change in the delivery promise is uncorrelated with the pretreatment delivery speed. Further, we run the DID with matching analysis and exploit the heterogeneous effects between new and existing customers. Our results are also robust to

alternative DID specifications and nonlinear model specifications, which are available in Online Appendix B.

8.1. Parallel Trends

The key identification assumption for the DID strategy is the parallel trends assumption, which is that before the policy change, the treatment and control groups would follow the same time trend. We perform empirical tests to confirm that there are no pretreatment differences in sales trends between the treatment and control cities. Because our treatments are continuous, we discretized the treatments and split them into a finite number of groups, as shown in Table 6. Figure 6 presents the time trends of the log orders. The two graphs show that both the slow and fast groups evolved in parallel with the control group before the policy change.

To formally test the parallel trend assumption, we estimate the following linear trend model for each outcome variable, restricting the sample to the pretreatment period:

$$Outcome_{it} = \alpha_i + \tau_t + \beta_1 Fast_i \times Week_t + \beta_2 Slow_i \times Week_t + e_{it}, \quad (4)$$

where α_i is the city-level fixed effect, τ_t is the week fixed effect, and $Week_t$ is the number of weeks at time t since the beginning of the sample period. The coefficients β_1 and β_2 capture the trend differences across treatment and control groups. The estimation results are reported in Table 7. The interaction terms are all insignificant, suggesting that the parallel trend assumption holds for our analysis.

8.2. Treatment Uncorrelated with Actual Delivery Speed Change

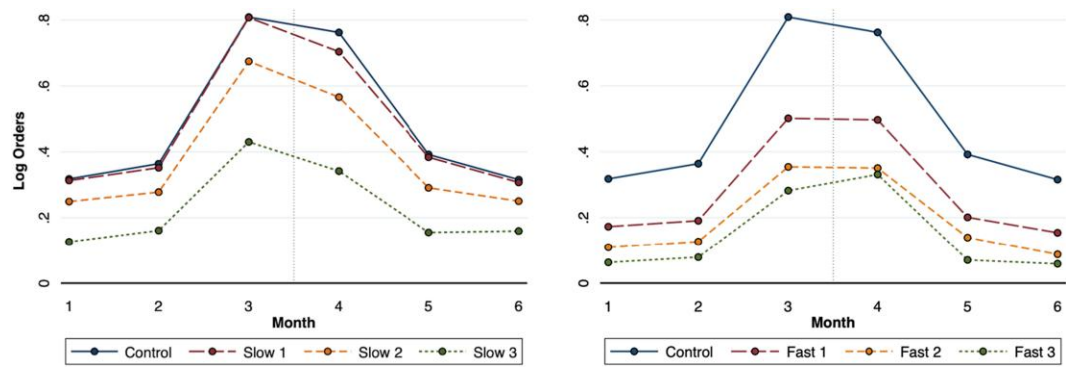
The DID identification requires the treatment assignment to be uncorrelated with the time-varying characteristics of cities (Angrist and Pischke 2008, Wooldridge 2010). Given our experiment design and treatment definition $\Delta d = d^{old} - d^{new}$, it is potentially possible for Δd to capture some past changes in the actual delivery speed.

In our context, we need to ensure that the treatment only changes the information aspect of delivery without being correlated to any potential changes in the physical delivery speed. This condition is likely to hold for the following reasons. Recall that all shipping is outsourced to third-party carriers, and the retailer did not alter its physical delivery setup in the six-month time window around the policy change. Within such a time period, it is unlikely that the offline infrastructure and physical delivery would change significantly. However, this does not

Table 6. Discretizing the Treatments

Group	Slow 3	Slow 2	Slow 1	Control	Fast 1	Fast 2	Fast 3
Δ Delivery Promise	(−6, −2.5]	(−2.5, −1.5]	(−1.5, −0.5]	(−0.5, 0.5)	[0.5, 1.5)	[1.5, 2.5)	[2.5, 6)

Figure 6. (Color online) Time Trends of Log Orders



Notes. These two graphs plot the time trends of the discretized treatment groups and the control group. Although our data have a weekly frequency, to smooth out fluctuations we average and plot log orders for each month.

mean that there is no variation in the physical delivery speed, as the actual delivery processes can be affected by other random factors, such as weather conditions and capacity. Nonetheless, as long as the change in delivery promise is independent of the change in actual delivery, we can still obtain an unbiased causal estimate. In other words, we need to rule out the possibility that the change in delivery promise is confounded with a potential change in physical delivery.

We directly test this assumption by regressing the change in delivery time on the change in delivery promise for the treatment group with an aggressive delivery promise and the treatment group with a conservative delivery promise separately:

$$\Delta \text{Delivery Time}_i = c + \beta \Delta \text{Delivery Promise}_i + e_i, \tag{5}$$

where $\Delta \text{Delivery Time}_i$ is the change in delivery time measured by days in city i before and after the policy and $\Delta \text{Delivery Promise}_i$ is the change in delivery promise in city i . Table 8 shows that both β_{Slow} and β_{Fast} are not significant. Figure 7 further illustrates that different groups' time trends of the actual delivery speed evolve in a parallel manner. In other words, the physical delivery speed difference between the treatment and control groups did not statistically significantly change after the

treatment. Therefore, all these empirical results confirm that the irrelevance assumption is supported by the data.

8.3. Treatment Uncorrelated with Pretreatment Delivery Speed

A key consideration in our empirical design is that the change in the delivery promise shall be uncorrelated with the pretreatment delivery speed. This design aims to rule out the possibility that our treatment enables a city with a previously speedy delivery (e.g., a city that is closer to the vendor) to get a faster speed promise after the policy change. In other words, we want to make sure that the speed promise change is exogenous to the pretreatment shipping speed. To empirically validate this condition, we test the correlation between the change in delivery promise and the pretreatment delivery speed at the city level:

$$\Delta \text{Delivery Promise}_i = c + \beta \text{Delivery Time}_i + \gamma X_i + e_i, \tag{6}$$

where $\Delta \text{Delivery Promise}_i$ is the change in delivery speed promise in city i ; Delivery Time_i is the average delivery time measured by days in city i before the policy change; and X_i is a vector of control variables that capture city-

Table 7. Parallel Trends Test

	Log orders (1)	Profit (2)	Log order value (3)	Log item price (4)	Log shipping expense (5)
Fast × Week	0.0003 (0.0005)	−13.94 (13.68)	−0.0038 (0.0035)	−0.0064 (0.0040)	−0.0031 (0.0027)
Slow × Week	0.0004 (0.0003)	−0.39 (10.77)	0.0001 (0.0014)	0.0010 (0.0016)	0.0001 (0.0012)
City FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Observations	85,080	85,080	22,063	22,063	22,063
R ²	0.005	0.003	0.004	0.002	0.006

Notes. This table reports the estimated coefficients and cluster-robust standard errors (in parentheses) in Equation (4). The coefficients for the number of orders, profit, the purchase value per order, the average price per item, and consumers' expenditure in shipping are presented in columns (1)–(5), respectively. The standard errors are clustered at the city level. FE stands for fixed effect.

Table 8. Treatment Uncorrelated with Actual Delivery Speed Change

	Coefficient value	Standard error	p-value	R ²	Observations
β_{Slow}	0.0224	0.0384	0.560	0.0001	4,150
β_{Fast}	−0.0265	0.1023	0.795	0.0001	960

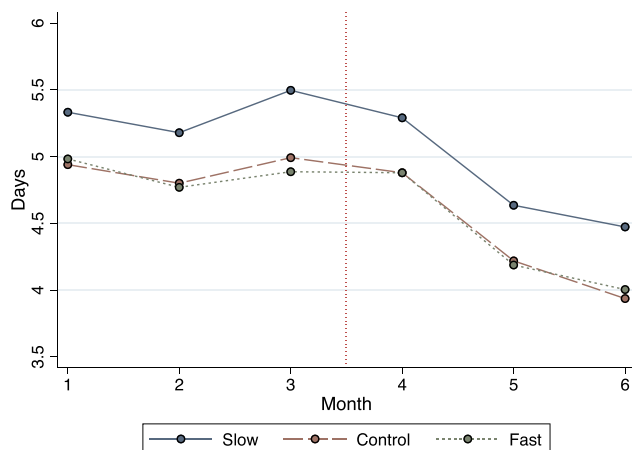
Notes. This table reports the estimated coefficients and robust standard errors (in parentheses) in Equation (5). The change in delivery promise is uncorrelated with the change in delivery speed. The low R^2 indicates that only 0.01% of the observed variations in $\Delta\text{Delivery Time}_i$ can be attributed to variations in $\Delta\text{Delivery Promise}_i$.

level characteristics, including city size, average income, population density, and poverty rates. To control for these characteristics, we link our city-level data to the 2010 census data,¹³ and more than 60% of cities in our sample are successfully matched. Table 9 shows that the change in delivery promise is uncorrelated with the pretreatment delivery speed. These empirical results validate our empirical design.

8.4. DID with Matching

For the new disclosure policy since Collage varied the delivery speed information across cities based on city characteristics, the treatment was not completely randomly assigned to different cities. To overcome this hurdle, we employ a DID analysis with matching (Fisher et al. 2019, Cui et al. 2020, Calvo et al. 2023), a common approach used to obtain a balanced sample that allows us to make an “apples-to-apples” comparison.

We run the analysis separately for the treatment group with faster delivery promises and the treatment group with slower delivery promises. We match cities in the treatment group and cities in the control groups that are

Figure 7. (Color online) Time Trends of Actual Delivery Speed

Notes. This graph shows the time trends of the actual delivery speed for both the treatment and control groups. Despite the presence of noticeable seasonality, which led to a decrease in the average delivery time during the sampled period, it is evident that the gap between the treatment and control groups remained consistent and unaffected.

Table 9. Treatment Uncorrelated with Pretreatment Delivery Speed

	Coefficient value	Standard error	p-value	R ²	Observations
β	−0.0123	0.0124	0.560	0.0204	4,455

Notes. This table reports the estimated coefficients and robust standard errors (in parentheses) in Equation (6). The change in delivery promise is uncorrelated with the pretreatment average delivery speed.

similar in terms of demographic characteristics—that is, city size, average income, population density, and poverty rates—but receive different treatment. We perform a one-to-one matching with replacement using the Mahalanobis distance (Ho et al. 2007); for each city in a treatment group, we find a “twin” city in the control group with the most similar feature vector. The matching process yields a subsample of 486 cities in the fast group matched with 389 cities in the control group and a subsample of 2,702 cities in the slow group matched with 1,028 cities in the control group. Table 10 shows that the treatment and control groups are balanced across key observable attributes.

We perform the DID analysis in Equation (1) using the matched samples. Table 11 reports the estimation results of the sample matched for the fast group, and Table 12 reports the estimation results of the sample matched for the slow group. The results are highly consistent with the estimations without matching in Table 3. A few coefficients become marginally insignificant, mainly because of the reduced sample size. The results become more significant when we increase the sample size by allowing for a one-to- N matching.

8.5. New Customers vs. Existing Customers

In this section, we study responses from new and existing customers. First, we report the customer composition in Figure 8(a). We can see that close to 80% of orders come from new customers who made their first purchase during our sample period, and the remaining orders are from existing customers who had made their first purchase prior to our sample period. Figure 8(b) further shows that nearly 40% of the existing customers shopped only once in the past, and most of them had purchased less than 10 times from the retailer.

Next, we empirically quantify the potential heterogeneous responses in purchasing behaviors between new and existing customers. We perform the DID analysis with existing customers’ order percentage as a moderator. Specifically, we first calculate the percentage of orders from existing customers, $PctExist_i$, for each city i , and we then estimate the following model:

$$\begin{aligned} Outcome_{it} = & \alpha_i + \tau_t + (\beta_1 + \beta_2 PctExist_i) Fast_i \times After_t \\ & + (\beta_3 + \beta_4 PctExist_i) Slow_i \times After_t + e_{it}, \end{aligned} \quad (7)$$

Table 10. Summary Statistics: Balanced Sample After Matching

Variable	Before matching			Fast matched		Slow matched	
	Fast	Control	Slow	Fast	Control	Slow	Control
Orders	1.00	1.75	1.69	1.00	1.06	1.69	1.63
Profit (\$)	81.33	142.16	133.70	81.33	87.32	133.70	132.11
Order value (\$)	106.40	106.25	103.55	106.40	108.71	103.55	105.81
Price (\$)	84.66	82.63	81.41	84.66	85.22	81.41	83.05
Shipping expenditure (\$)	13.16	13.11	13.00	13.16	13.37	13.00	13.13
Number of cities	486	1,270	2,702	486	389	2,702	1,028

where the outcome variable takes its value from logged sales, profit, logged purchase value, logged price of purchased products, and logged expenditure on shipping. If existing customers respond to the speed information differently from new customers, we would expect the treatment effects in cities with a higher percentage of repeat orders to be different from those in cities with a lower percentage of repeat orders; that is, β_2 and β_4 would be significant.

Table 13 reports the estimation results. It can be seen that the coefficients of $Slow \times After \times PctExist$ and $Fast \times After \times PctExist$ are mostly insignificant, which implies that there is no difference between new and existing customers' purchasing behaviors when changing the shipping speed promise.

To quantify the potential heterogeneous responses in returns and satisfaction between new and existing customers, we augment Equation (2) with a new variable $Existing_i$ and estimate the following model:

$$\begin{aligned} Outcome_{ijt} = & \alpha_i + \tau_t + (\beta_1 + \beta_2 Existing_j) Fast_j \times After_t \\ & + Existing_j + (\beta_3 + \beta_4 Existing_j) Slow_j \\ & \times After_t + \gamma' X_j + e_{ijt}, \end{aligned} \tag{8}$$

where $Existing_i$ is a dummy variable indicating whether individual i is an existing customer. The two interaction terms, $Fast_j \times After_t \times Existing_i$ and $Slow_j \times After_t \times Existing_i$, capture the different responses of existing customers compared with new customers.

Table 14 reports the estimation results. First, we find that existing customers are less likely to return and give higher NPS scores according to columns (1), (2), and (5). This is reasonable as there is a strong self-selection effect; customers who are better matched with Collage's products and services will choose to revisit and make repeat purchases. We also find significant differences between new and existing customers' return behaviors in response to a speed promise change. The coefficient of $Slow \times After \times Existing$ is significant in column (1), and the coefficient of $Fast \times After \times Existing$ is significant in column (2). Both the linear and logit models indicate that existing customers are more sensitive to a faster delivery speed promise but less sensitive to a slower promise. These findings suggest that the retailer should be more cautious about making aggressive promises to existing customers.

9. Conclusions

By studying a new shipping disclosure policy for Collage.com, this paper takes one of the first steps toward empirically quantifying the economic impact of disclosing aggressive or conservative shipping estimates to customers. We demonstrate that the online retailer can benefit from managing the disclosure of delivery speed information. It is an effective way to boost sales, profits, and customer retention as well as to reduce product returns. The results are robust to alternative DID model specifications, the DID with matching method, and non-linear model specifications. The findings highlight the

Table 11. Effects on Sales and Spending Using Matched Sample (Fast)

	Log orders (1)	Profit (2)	Log order value (3)	Log item price (4)	Log shipping expense (5)
<i>Fast</i> × <i>After</i>	0.0115* (0.0066)	56.8 (161.7)	0.0380** (0.0193)	0.0305 (0.0233)	0.0318* (0.0168)
City FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Observations	21,000	21,000	7,937	7,937	7,937
R ²	0.256	0.181	0.022	0.026	0.038

Notes. This table reports the estimated coefficients and cluster-robust standard errors (in parentheses) in Equation (1) with the matched sample. For each city from the fast group, we match it with a closest city from the control group with replacement according to the Mahalanobis distance. The coefficients for the number of orders, profit, the purchase value per order, the average price per item, and consumers' expenditure in shipping are presented in columns (1)–(5), respectively. The standard errors are clustered at the city level. FE stands for fixed effect.

* $p < 0.1$; ** $p < 0.05$.

Table 12. Effects on Sales and Spending Using Matched Sample (Slow)

	Log orders (1)	Profit (2)	Log order value (3)	Log item price (4)	Log shipping expense (5)
<i>Slow × After</i>	−0.0134*** (0.0027)	−413.3*** (92.1)	−0.0313*** (0.0062)	−0.0404*** (0.0078)	−0.0283*** (0.0052)
City FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Observations	89,520	89,520	42,137	42,137	42,137
R^2	0.303	0.122	0.023	0.016	0.050

Notes. This table reports the estimated coefficients and cluster-robust standard errors (in parentheses) in Equation (1) with the matched sample. For each city from the slow group, we match it with a closest city from the control group with replacement according to the Mahalanobis distance. The coefficients for the number of orders, profit, the purchase value per order, the average price per item, and consumers' expenditure in shipping are presented in columns (1)–(5), respectively. The standard errors are clustered at the city level. FE stands for fixed effect.

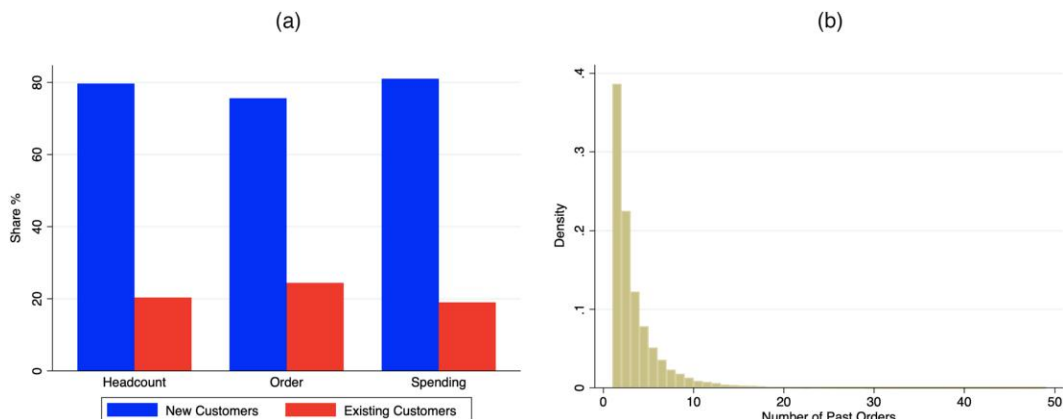
*** $p < 0.01$.

trade-offs of promising delivery speed information; executing an aggressive disclosure can increase sales at the cost of more product returns and a lower retention rate in the future, and executing a conservative disclosure can improve customer future retention at the cost of less sales.

The findings of our study can be generalized to other retail contexts. Given that Collage is one of the largest retailers in the photo product market, it is large in both scale and scope. Our research—which involved hundreds of product SKUs and millions of consumers and transactions—represents a rich context. In addition, consumers' decision processes in our study are very similar to those in other online shopping experiences. After a customer chooses a product and types in her zip code, Collage will provide her an estimated date by which time the order will be delivered, the very same experience that customers have with Amazon. Nevertheless, it is possible that Collage's customers have different time

sensitivity than the customers of other online retailers, such as Amazon, which is a quantitative rather than a qualitative difference in consumer behavior. Companies can apply our managerial insights to facilitate their decisions but would still need to reestimate the effects in their own contexts. Therefore, our managerial implications can be generalized to and guide information strategies for other online retailers.

Because of various practical constraints, our study was not based on a fully randomized experiment. Although our experiment incorporated some randomness in assigning the treatment, there could be concerns that our treatment might be correlated with city-level characteristics. To eliminate these concerns, we employed the DID method and verified its parallel trends assumption. However, DID only identifies the treatment effect on those who were selected for treatment, and hence, the estimated causal effect does not necessarily apply to the entire population, potentially limiting the generalizability of

Figure 8. (Color online) New and Existing Customers

Notes. (a) Decomposition by customer type. (b) Number of past orders of existing customers. Panel (a) shows the decomposition of new and existing customers. We can see that around 80% of customers are new customers who contributed 75% of the total orders and 81% of the total spending. Panel (b) shows that nearly 40% of existing customers shopped only once in the past few years. Most of them had made fewer than 10 purchases from the retailer.

Table 13. The Moderating Effects of Existing Customers on Sales and Spending

	Log orders (1)	Profit (2)	Log order value (3)	Log item price (4)	Log shipping expense (5)
<i>Fast × After</i>	0.0054 (0.0038)	99.0 (105.6)	0.0289* (0.0168)	0.0272 (0.0196)	0.0246* (0.0139)
<i>Slow × After</i>	−0.0051** (0.0020)	−252.4*** (59.1)	−0.0293*** (0.0069)	−0.0444*** (0.0098)	−0.0284*** (0.0058)
<i>Fast × After × PctExist</i>	0.0084 (0.0079)	357.0** (166.7)	0.0265 (0.0088)	0.0203 (0.0634)	−0.0211 (0.0487)
<i>Slow × After × PctExist</i>	−0.0003 (0.0048)	28.8 (90.8)	−0.0073 (0.0185)	0.0036 (0.0233)	−0.0126 (0.0160)
City FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Observations	170,160	170,160	60,748	60,748	60,748
R ²	0.243	0.093	0.022	0.015	0.047

Notes. This table reports the estimated coefficients and cluster-robust standard errors (in parentheses) in Equation (7). The coefficients for the number of orders, profit, the purchase value per order, the average price per item, and consumers’ expenditure in shipping are presented in columns (1)–(5), respectively. The standard errors are clustered at the city level. FE stands for fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

our findings. Therefore, we recommend randomized experiments whenever possible to ensure more robust and comprehensive results.

Our insights can shed further light on how retailers can tailor their information strategies. Given the uncertainty in physical delivery speed, there is always value in customizing the speed promised to customers. The customization depends on how sensitive customers are to faster and slower promises, which has been empirically estimated in our study. Based on these estimates,

retailers can employ the framework provided in Section 7 to determine the optimal promise. Specifically, retailers face tension between short-term and long-term profitability. When promising one day faster (slower), the marginal benefit (cost) is short-term profits, which are represented by a net revenue increase (decrease); the marginal cost (benefit) is long-term profits, which are represented by the lifetime value of lost (retained) customers. By carefully balancing the two forces, one can derive the optimal promise that maximizes retailers’

Table 14. The Moderating Effects of Existing Customers on Returns and Satisfaction

	Return		Satisfied		NPS
	(1) Linear	(2) Logit	(3) Linear	(4) Logit	(5) Linear
<i>Fast × After</i>	0.0005* (0.0003)	0.0731** (0.0341)	−0.0033 (0.0028)	0.0120 (0.0374)	0.0040 (0.0210)
<i>Slow × After</i>	−0.0006** (0.0002)	−0.0972*** (0.0367)	0.0023 (0.0038)	−0.0306 (0.0335)	0.0145 (0.0279)
<i>Existing</i>	−0.0011** (0.0005)	−0.2413*** (0.0925)	0.0062 (0.0071)	0.1519* (0.0805)	0.3698*** (0.0492)
<i>Fast × After × Existing</i>	0.0007 (0.0006)	0.1107** (0.0522)	0.0039 (0.0054)	0.0100 (0.0681)	−0.0383 (0.0363)
<i>Slow × After × Existing</i>	0.0006* (0.0003)	0.0864 (0.0645)	−0.0048 (0.0056)	−0.0612 (0.0523)	−0.0552 (0.0387)
Controls	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
Observations	175,647	170,554	13,719	15,635	13,719
R ²	0.012	0.055	0.059	0.077	0.077

Notes. This table reports the estimated coefficients and standard errors (in parentheses) in Equation (2). Regressions are run at the transaction level. The estimates for whether the order is returned using linear and logit models are presented in columns (1) and (2). The estimates for whether the customer is satisfied using linear and logit models are presented in columns (3) and (4). The estimates for the satisfaction score are presented in column (5). For linear models, city and week fixed effects are included, and standard errors are clustered at the city level. For nonlinear models, we use state-level fixed effects instead to avoid identification issues caused by the high dimensionality of city-level fixed effects. We report adjusted R^2 for linear models and pseudo- R^2 for nonlinear models. FE stands for fixed effect.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

profitability. Our quantitative analysis finds that by promising two days faster, Collage can increase its profits by more than 2%.

Companies competing in the online retailing space can leverage this study by tailoring their information disclosure policies as a function of individual characteristics. For example, retailers can use data-driven policies, such as causal machine learning, to prescribe the delivery estimates in light of the ever-changing context of each product, geographical region, and individual customer. With the rapid advancement in firms' capabilities to process a large amount of real-time consumer behavior data, we envision the ability of firms to optimize and share the delivery speed information in an instantaneous and personalized fashion.

The retailer that we partnered with has found value in and is ready to implement our recommendations. In the words of the retailer's CEO, this project "provides us insights that help us discuss and subsequently implement changes in our shipping policy, as well as test alternative shipping information models in our company." Considering the emphasis on delivery speed and stability, the significance of firms' information-sharing strategies has been intensified because of their ever-increasing impact on consumer decisions. We hope that this paper will serve as a stepping stone for future research to explore how to manage and optimize delivery speed information in various dimensions. For example, the retailer in our study promises delivery speed in the form of a point estimate (i.e., providing an exact date by which the order will arrive); future research could investigate other forms, such as an interval estimate.

Endnotes

¹ The key identification assumption of DID is parallel trends. Parallel trends may fail to hold because of unobservable time-varying confounders. We would like to point out that the semirandomness is not necessary nor sufficient to our DID identification strategy; however, it helps eliminate the potential unobservable time-varying confounders.

² The shipping fee policy of Collage.com is illustrated in Figure 16 in Online Appendix D. First, shipping is not free; even the standard ground shipping has a charge. Second, shipping is more costly for more expensive products.

³ There is no need to include $After_t$ in the regression because the new policy was implemented at the beginning of a week. All its effects are incorporated by τ_t .

⁴ Because some observations are zero, we actually use the $\log(1 + x)$ transformation.

⁵ Because the profit is negative for some observations, we cannot use the logged value here.

⁶ For returns and satisfaction, we can also perform the analysis at the transaction level. The results are similar.

⁷ Note that the number of observations in columns (3)–(5) of Table 3 is less than the number of observations in columns (1) and (2). This is because some cities did not have any transactions during some weeks, and as a result, the values of those three variables (i.e., the average order value, the average item price, and the shipping expense) are missing.

⁸ Because the outcome variables are logged, the percentage change in sales equals $e^{0.0073} - 1 = 0.73\%$. Similar transformations are used for order value, prices of purchased items, and shipping expenditure.

⁹ The percentages are calculated by $\Delta\text{profit}/\text{average profit}$, where the denominator is the average profit before the policy change.

¹⁰ In Table 4, the linear and logit models have different numbers of observations for two reasons. First, the linear models drop singleton observations because they bias the cluster-robust standard error. Second, the logit models may drop some observations when there are categorical control variables, such as the state and the product fixed effects. If there are no variations in the outcome variable within a state or a product category, the maximum likelihood estimator cannot identify the corresponding coefficients. Thus, all these observations will be dropped.

¹¹ The percentage is calculated by $\Delta\text{return rate}/\text{average return rate}$, according to which we get $(0.06/0.62) = 9.7\%$ and $(0.04/0.62) = 6.4\%$.

¹² The results are similar but quantitatively larger when we extend the time window from one to three years (i.e., using three-year retention instead of one-year retention).

¹³ The source is <https://www.census.gov/programs-surveys/decennial-census/decade.2010.html> (accessed December 2020).

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